An Adaptive Wrist-Worn Device that Learns Each Patient's Tremor Pattern to Suppress and Track Parkinson's Disease Rest Tremor in Daily Life

IBEHS 5P06 Capstone

Progress Report 1: Background, Motivation, and Needs

Team 19

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1 Background and Motivation

1.1 Problem context and engineering opportunity

Parkinson's disease affected more than 8.5 million people worldwide in 2019 [1]. In Canada, more than 120,000 people live with the disease; its estimated 2024 cost was $3.3 billion, approximately 90% borne by patients and care partners [2]. Classical Parkinsonian rest tremor occurs at 4–6 Hz [3], although postural and kinetic tremors also occur [4]. In a 2025 survey of 634 patient and care-partner respondents, tremor interfered with handwriting for 60%, keyboard use for 52%, and smartphone or tablet use for 47%; approximately half reported inadequate medication response [4].

Functional limitations, stigma and the invasiveness of some treatments motivate investigation of an additional non-invasive aid [5]. Symptom variability [6] and episodic clinical assessment [9] also support longitudinal home monitoring. Our initial technical focus is rest-tremor suppression and tracking while preserving voluntary movement. Improved eating, writing or phone use remains a broader outcome requiring separate evaluation; it cannot be inferred from reduced rest tremor.

Cala kIQ offers a comparison for temporary relief of postural and kinetic hand tremor in Parkinson's disease [7]. Its 2023 U.S. Veterans Affairs prices of US$3,200 plus US$1,880 annually for bands are a historical cost benchmark, not current Canadian pricing [8]. Its indication differs from our initial rest-tremor focus.

1.2 Stakeholder engagement and intended use

In September 2026, software co-supervisor Dr. Spencer Smith emphasized safe non-human testing, affordability and longitudinal monitoring given the team's limited mechanical expertise. Reviewer Mohamad Mirhoseininejad highlighted technical and patient-reported metrics, symptom adaptation and preservation of voluntary movement. Biomedical co-supervisor Vincent Leung approved the project, subject to clarification of ethics requirements for stakeholder involvement.

Parkinson Canada's Calaina Brooke, PhD, agreed to connect the team with its new-products and research group; direct user validation remains pending. Clinical outreach to Hamilton Health Sciences / McMaster continues. Intended users are independently living adults with Parkinson's hand tremor at home and in the community, with care partners assisting as needed and clinicians reviewing symptom summaries. Wear duration, assistance needs and report content require stakeholder input.

1.3 Preliminary concept and next steps

The candidate concept combines wrist motion sensing, predictive control of a magnetorheological damper, and a companion app for tremor trends and user-entered context. These mechanisms remain provisional. Next, we will evaluate prediction on suitable public tremor data and model damping before developing a mechanical tremor rig. This sequence addresses the feasibility and safety concerns raised by supervisors while allowing the monitoring component to progress independently.

2 User Needs and Project Statement

2.1 Outcomes and Mechanisms

Table 1 separates desired outcomes from the candidate mechanisms. Needs reflect literature and stakeholder discussions; team-derived needs and acceptable performance thresholds require validation with intended users before detailed requirements are finalized.

Table 1. Preliminary user needs and their evidence or validation status.

ID	*The user needs to*	*Basis and status*
N1	Perform daily tasks with less tremor interference.	Literature [4]; task benefits require separate evaluation.
N2	Obtain relief suited to changing individual symptoms.	Variability [6]; reviewer feedback.
N3	Move voluntarily, including typing, without interference.	Reviewer feedback; tolerable resistance unresolved.
N4	Use the aid safely during normal use and foreseeable failures.	Dr. Smith; unsupervised use remains provisional.
N5	Put on, wear and operate the aid comfortably.	Team-derived; wear duration and assistance unresolved.
N6	Share useful tremor trends and context with clinicians.	Monitoring literature [9]; Dr. Smith; content unvalidated.
N7	Afford purchase, maintenance and replacement costs.	Financial burden [2]; acceptable cost unresolved.
N8	Control access to personal health data.	Team-derived privacy requirement.

2.2 Expected impact

A successful design could support independence and reduce visible-tremor stigma [5]. Care-partner workload could decrease, although setup and maintenance may add demands. Safety and comfort must be balanced against suppression, and affordability must include ongoing costs. Monitoring could complement clinic assessments [9] if summaries are useful and sharing remains user-controlled. Appearance and visibility preferences require consultation. Durability and rechargeable components are environmental objectives whose benefits require lifecycle evaluation.

2.3 Risks and evidence of progress

Key barriers are mechanical expertise, actuation delay, movement restriction and access to user feedback. Public datasets, simulation and a proposed tremor rig provide an initial non-human test path; component selection and mechanical advising will address integration risks. Before involving people with Parkinson's disease, we will confirm institutional ethics requirements [10] and any applicable device-testing authorization [11]. Privacy obligations will be assessed against the intended data flows.

Progress will be measured through detection agreement with reference labels, prediction error against a simple baseline, and rig comparisons with damping disabled, non-predictive damping and predictive damping. Tests will assess tremor amplitude and resistance to simulated voluntary movement. Stakeholder feedback and initial results will guide acceptance criteria; later permitted usability testing would assess task completion, assistance, comfort and satisfaction.

People living with Parkinson's disease need a safe, affordable way to reduce hand-tremor interference in daily life, accommodate changing symptoms, preserve voluntary movement and share an objective record of tremor over time.

References

[1] World Health Organization, "Parkinson disease," Fact sheet, 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/parkinson-disease

[2] Parkinson Canada, "Understanding the reality," 2025. [Online]. Available: https://www.parkinson.ca/the-reality/

[3] W. Chen, F. Hopfner, J. S. Becktepe, and G. Deuschl, "Rest tremor revisited: Parkinson's disease and other disorders," *Transl. Neurodegener.*, vol. 6, art. 16, 2017.

[4] R. A. Hauser *et al.*, "Burden of tremor in Parkinson's disease: A survey study," *J. Parkinson's Dis.*, vol. 15, no. 3, pp. 541–551, 2025, doi: 10.1177/1877718X251329328.

[5] B. E. K. S. Swinnen, A. W. G. Buijink, A. J. Espay, and R. M. A. de Bie, "Tremor in Parkinson's disease," *Semin. Neurol.*, 2025, doi: 10.1055/a-2768-3003.

[6] J. E. Thorp, P. G. Adamczyk, H.-L. Ploeg, and K. A. Pickett, "Monitoring motor symptoms during activities of daily living in individuals with Parkinson's disease," *Front. Neurol.*, vol. 9, art. 1036, 2018.

[7] Anthem, "DME.00049 External upper limb stimulation for the treatment of tremors," Medical policy, 2026. [Online]. Available: https://www.anthem.com/medpolicies/abc/active/mp_pw_e001873.html

[8] Cala Health, "VA Cala kIQ ordering process for prosthetics department," 2023. [Online]. Available: https://calahealth.com/wp-content/uploads/2023/09/SD-20010-SD-20011-MM-20036-VA-Website-Download.pdf

[9] M. Sica *et al.*, "Continuous home monitoring of Parkinson's disease using inertial sensors: A systematic review," *PLoS ONE*, vol. 16, no. 2, art. e0246528, 2021.

[10] Canadian Institutes of Health Research, Natural Sciences and Engineering Research Council of Canada, and Social Sciences and Humanities Research Council, *Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans – TCPS 2 (2022)*. Ottawa, ON, Canada, 2022.

[11] Health Canada, "Guidance on the registration of clinical trials and public disclosure of results: Appendices." [Online]. Available: https://www.canada.ca/en/health-canada/services/drugs-health-products/drug-products/applications-submissions/guidance-documents/guidance-registration-clinical-trials-public-disclosure-results/appendices.html